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Combine Dataframes and Re-Structured

Data re-structured and summarized

Create a Table for HTML Output

Data Dictionary

MILESTONE 4 STARTS HERE

Please submit an Rmd or Qmd and publish an html file on RPubs with the following:

1. Final datasets for creation of visualization
   - Join all datasets together
   - Calculate any remaining data elements needed for analysis
   - Show code used to create joined dataset, but please do not print full data frame output (showing data structure with `str()` is okay)
2. Visualizations (at least one per group member)
   - One print quality table as requested in scenario
   - One print quality plot or chart as requested in scenario
   - For groups of 3, one additional print quality table or plot of your choice (can support the requested data in the scenario, or answer a different question using the same data sources)
3. For each visual, include
   - Code used to generate visual

- Legend (if necessary)
- 1-2 sentence interpretation
- NOTE:
  - Please make sure the visual can stand-alone, meaning it includes enough information in title, legend, and footnote so that a person who sees only the visualization could understand what is being presented.
  - Please also make sure column names, axis labels, and any other labels are meaningful and not just the name of the variable (ex: "County" rather than "county_name")

4. Html file that is professionally prepared for presentation and published to RPubs

- Only the necessary information is in the output (e.g., suppress entire data frame outputs but showing data structure with `str()` is okay.
- Code used to import and clean data (milestones 2 and 3) can be included in the rmd/qmd but please use the option include=FALSE in order to prevent the code and any output from. Code chunk would look like `{r chunk_name, include=FALSE}`
- Utilize `warning=FALSE` and `message=FALSE` to suppress any warnings/messages that may be displaying from any chunks
- For this milestone please make sure code for visualizations are included in the final file. Show your work by "echoing" code used in Rmd/Qmd file to create your tables and graphs. "'{r, echo = TRUE} in code chunk preferences.
- Use headers and sub headers to create an organized document
- Make sure you commit and push your rsconnect folder! Read more details here.Links to an external site.

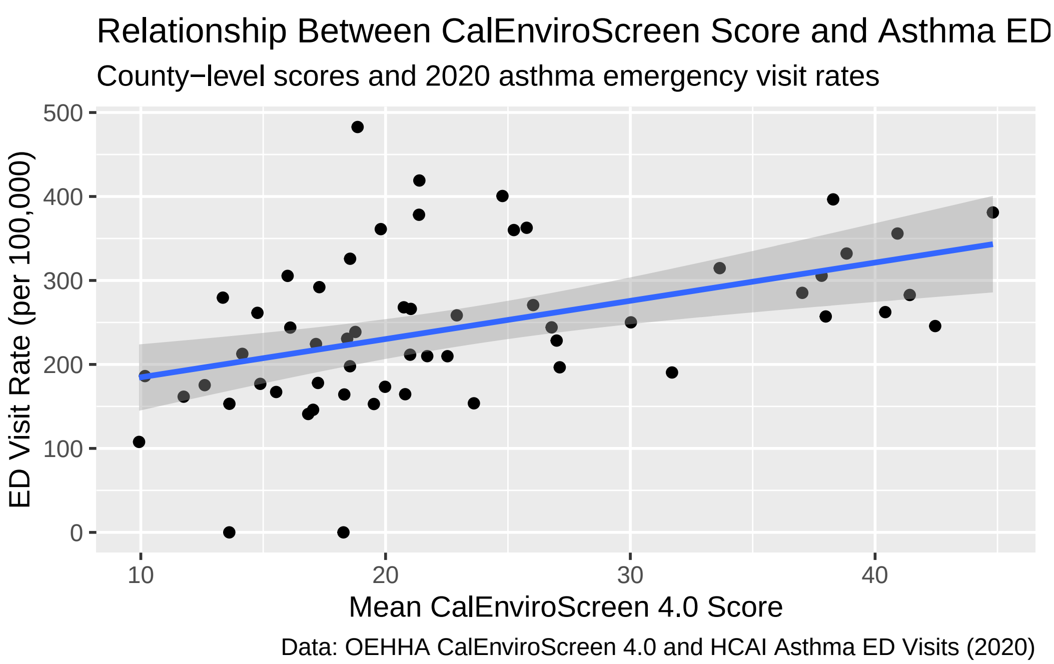
Final datasets for creation of visualization

```
df_combined <- df_combined %>%
  mutate(
   ces_category = case_when(
      mean_ces_4_0_score < 20 ~ "Low",
      mean_ces_4_0_score < 30 ~ "Moderate",
      TRUE ~ "High"),
    high_poverty = if_else(median_poverty > median(median_poverty),
                           "High poverty", "Low poverty"),
```

```
    poverty_pollution_ratio = median_poverty / median_pm2_5
  )
```

```
ggplot(df_combined, aes(x = mean_ces_4_0_score,
                        y = ed_visits_per_100k)) +
  geom_point() +
  geom_smooth(method = "lm") +
  labs(
    title = "Relationship Between CalEnviroScreen Score and Asthma ED Visit Rates",
    subtitle = "County-level scores and 2020 asthma emergency visit rates",
    x = "Mean CalEnviroScreen 4.0 Score",
    y = "ED Visit Rate (per 100,000)",
    caption = "Data: OEHHA CalEnviroScreen 4.0 and HCAI Asthma ED Visits (2020)"
  )
```

```
`geom_smooth()` using formula = 'y ~ x'
```



a) County-level summary table

- **Purpose:** Show key county metrics side by side for easy comparison.
- **Columns:**

  - County
  - Mean CES score
  - CES category
  - Median PM2.5
  - Median diesel PM
  - Median poverty
  - Asthma ED visit rate (age-adjusted)
  - ED visits per 100k

- **Interpretation:** Lets you see which counties have both high environmental exposure and high asthma rates. Could help prioritize intervention targets.
- **R example (using `kable` from `knitr` or `gt` for more polished tables):**

```r
df_combined %>%
  select(county, mean_ces_4_0_score, ces_category, median_pm2_5, median_diesel_pm, median_pov
  arrange(desc(age_adjusted_ed_visit_rate)) %>%
  kable(caption = "County-level Environmental Measures and Asthma ED Rates (2020)")
```

Table 3: County-level Environmental Measures and Asthma ED Rates (2020)

county	mean_ces_4_0_score	ces_category	median_pm2_5	median_diesel_pm	median_poverty	age_adjusted_ed_visit_rate
Modoc County	18.856868	Low	4.053603	0.0092467	48.05	57.3
Lake County	21.380735	Moderate	3.685701	0.0150440	44.40	46.8
Plumas County	15.997295	Low	7.322027	0.0057812	28.30	42.2
Solano County	24.775746	Moderate	8.545621	0.1218199	23.10	41.7
Del Norte County	21.366216	Moderate	5.746661	0.0231158	43.80	40.6
Mendocino County	19.804257	Low	6.603766	0.0118368	40.00	39.8
San Joaquin County	38.288198	High	10.810312	0.1397718	39.00	38.6
Merced County	44.813759	High	11.950705	0.0932147	47.20	37.1
San Benito County	25.765146	Moderate	5.015597	0.0870067	25.00	36.5

county	mean_ces_4_0_score	ces_category	median_pm2_5	median_diesel_pm	median_poverty	age_adjusted_ed_visit_rate
Humboldt County	18.550177	Low	6.173489	0.0218621	38.40	36.0
Sacramento County	25.240882	Moderate	8.781627	0.1114184	30.55	35.7
Fresno County	40.915469	High	13.539645	0.1088067	47.55	34.0
Lassen County	17.293615	Low	4.688027	0.0048199	33.50	33.4
Mono County	13.352221	Low	3.339250	0.0013549	30.10	33.3
Stanislaus County	38.836013	High	11.205470	0.1591893	37.65	32.8
Amador County	20.744978	Moderate	8.239071	0.0121861	23.10	31.2
San Bernardino County	33.652742	High	11.724374	0.1316175	38.60	31.1
Madera County	37.816074	High	12.181452	0.0615622	35.90	30.8
Calaveras County	16.106966	Low	8.435455	0.0074388	26.90	30.4
Tuolumne County	18.766944	Low	8.237933	0.0086210	26.25	28.6
Trinity County	14.146272	Low	4.020512	0.0017212	39.90	28.5
Inyo County	14.764710	Low	4.135388	0.0021986	26.35	28.3
Tehama County	26.031105	Moderate	7.216373	0.0335159	42.50	28.0
Kern County	37.025458	High	15.067304	0.0928947	51.90	27.7
Contra Costa County	21.030046	Moderate	8.779497	0.1511612	16.30	27.5
Kings County	41.416584	High	13.430797	0.0655328	48.00	27.1
Los Angeles County	37.984394	High	11.905900	0.1960163	33.10	26.2
Alameda County	22.908671	Moderate	8.705996	0.2604092	17.15	25.8
Imperial County	40.416070	High	8.971584	0.0653801	47.60	25.1
Yuba County	30.023819	High	8.493959	0.0647969	36.80	25.0
Riverside County	26.784122	Moderate	9.921096	0.1008070	34.30	24.9

county	mean__ces_4_0_score	ces_category	median__pm2_5	median__diesel_pm	median__poverty	age_adjusted_ed_visit_rate
Shasta County	17.157221	Low	8.322866	0.0524080	33.85	24.8
Siskiyou County	18.546824	Low	3.387377	0.0087358	39.55	24.8
Butte County	21.703218	Moderate	8.446836	0.0702956	36.00	24.2
Colusa County	26.989905	Moderate	7.775336	0.0349152	37.50	24.0
Tulare County	42.455609	High	13.703033	0.0857922	53.65	23.8
Mariposa County	17.237830	Low	8.152019	0.0012921	35.45	22.8
Yolo County	22.528640	Moderate	8.787217	0.1075726	36.20	22.8
Napa County	18.430822	Low	9.265583	0.0963095	21.70	22.7
Nevada County	12.609442	Low	6.646795	0.0196855	25.60	21.6
Monterey County	21.003304	Moderate	5.503163	0.0779020	33.45	21.3
El Dorado County	10.169962	Low	7.631881	0.0204343	20.35	20.6
Glenn County	27.114924	Moderate	7.511624	0.0529494	40.30	19.5
Sonoma County	14.880642	Low	7.383509	0.0817708	21.30	18.8
Sutter County	31.702736	High	9.107645	0.0915673	39.00	18.8
San Luis Obispo County	13.619140	Low	7.414676	0.0465301	23.40	18.3
San Diego County	19.979314	Low	9.647294	0.1394067	25.00	17.5
Ventura County	20.801510	Moderate	8.981781	0.0990142	20.10	17.4
Santa Cruz County	15.530385	Low	6.127600	0.0828977	26.25	17.3
Placer County	11.750681	Low	7.904845	0.0733756	18.15	16.7
San Francisco County	18.313167	Low	8.616607	0.4834492	18.00	16.6
Orange County	23.608586	Moderate	11.708609	0.1519407	20.10	15.9

county	mean_ces_4_0_score	ces_category	median_pm2_5	median_diesel_pm	median_poverty	age_adjusted_ed_visit_rate
Santa Barbara County	19.523026	Low	7.663210	0.1116346	25.80	15.5
Santa Clara County	17.039068	Low	8.271690	0.1864493	14.55	14.5
San Mateo County	16.839289	Low	8.425126	0.1593324	12.75	14.0
Marin County	9.927999	Low	8.381069	0.0532390	12.90	11.7
Alpine County	13.615164	Low	3.054073	0.0026241	38.90	0.0
Sierra County	18.279114	Low	6.421772	0.0023986	31.70	0.0

b) Stratified summary table Purpose: Compare asthma rates by poverty level or CES category. Example: Mean asthma ED rate by CES category:

   Interpretation: Shows if high CES or high poverty counties tend to have higher asthma ED rates.

```
df_combined %>%
  group_by(ces_category) %>%
  summarise(mean_ed_rate = mean(age_adjusted_ed_visit_rate, na.rm = TRUE),
            mean_pm2_5 = mean(median_pm2_5, na.rm = TRUE),
            mean_poverty = mean(median_poverty, na.rm = TRUE))
```

```
# A tibble: 3 x 4
  ces_category mean_ed_rate mean_pm2_5 mean_poverty
  <chr>               <dbl>      <dbl>        <dbl>
1 High                 29.1      11.7          42.8
2 Low                  23.4       6.78         28.0
3 Moderate             28.5       7.84         30.8
```

```
df_combined %>%
  group_by(county, ces_category) %>%
  summarise(mean_ed_rate = mean(age_adjusted_ed_visit_rate, na.rm = TRUE),
            mean_pm2_5 = mean(median_pm2_5, na.rm = TRUE),
            mean_poverty = mean(median_poverty, na.rm = TRUE)) %>%
  ungroup() %>%
  arrange(desc(mean_ed_rate))
```

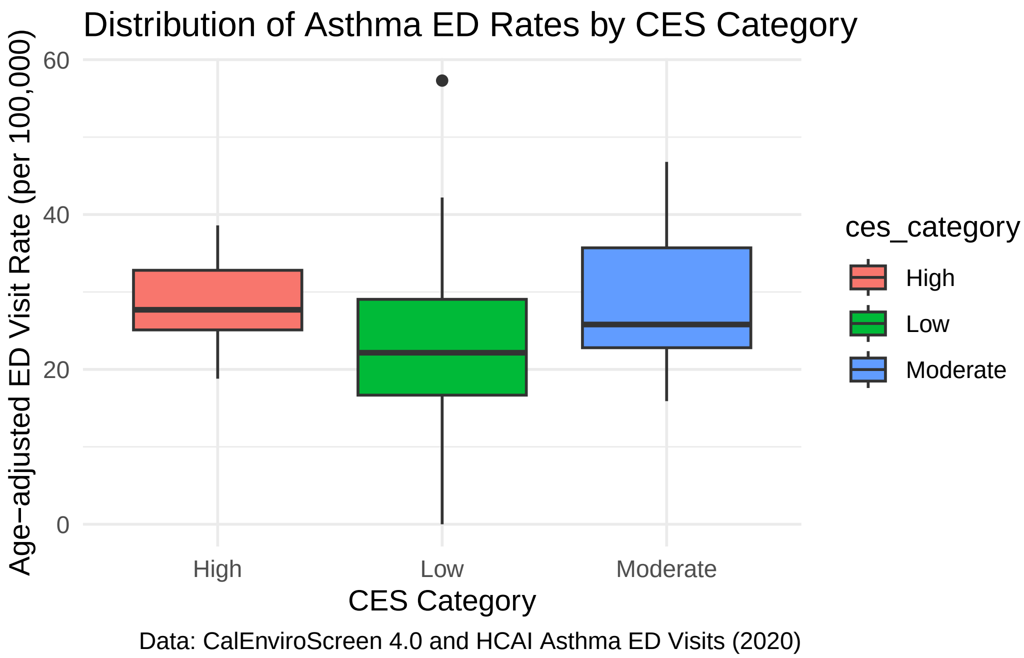
```
`summarise()` has grouped output by 'county'. You can override using the
`.groups` argument.
```

```
# A tibble: 58 x 5
   county             ces_category mean_ed_rate mean_pm2_5 mean_poverty
   <chr>              <chr>               <dbl>      <dbl>        <dbl>
 1 Modoc County       Low                  57.3       4.05         48.0
 2 Lake County        Moderate             46.8       3.69         44.4
 3 Plumas County      Low                  42.2       7.32         28.3
 4 Solano County      Moderate             41.7       8.55         23.1
 5 Del Norte County   Moderate             40.6       5.75         43.8
 6 Mendocino County   Low                  39.8       6.60         40
 7 San Joaquin County High                 38.6      10.8          39
 8 Merced County      High                 37.1      12.0          47.2
 9 San Benito County  Moderate             36.5       5.02         25
10 Humboldt County    Low                  36         6.17         38.4
# i 48 more rows
```

Boxplot of asthma ED rates by CES category

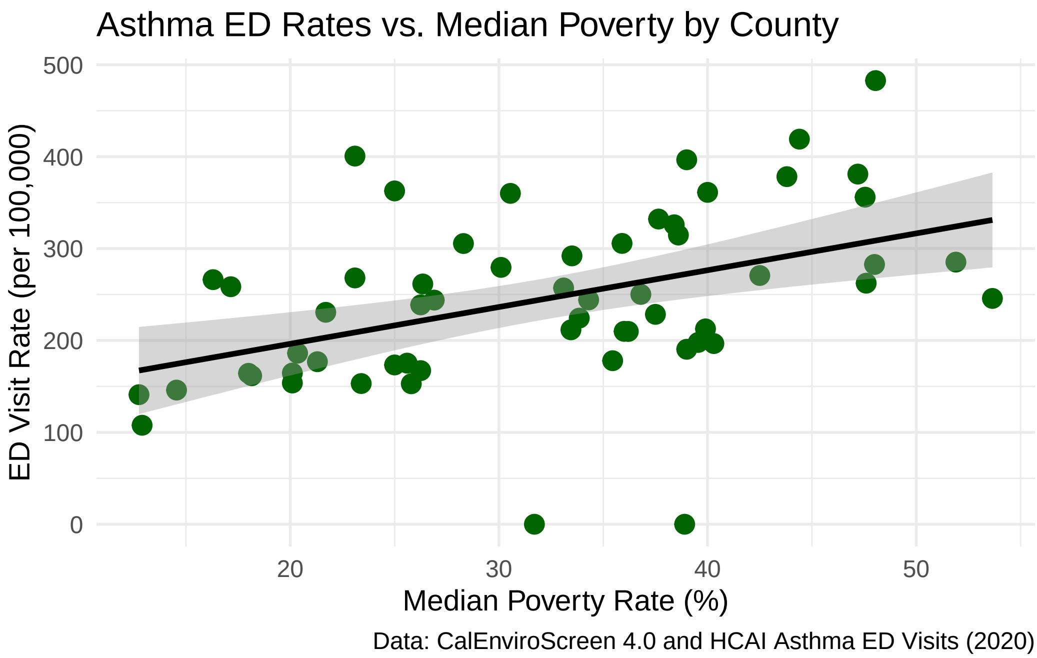
- **Purpose:** Visually compare distributions across "Low," "Moderate," and "High" CES counties.
- **Interpretation:** Highlights disparities in asthma rates across environmental burden categories.

```r
ggplot(df_combined, aes(x = ces_category, y = age_adjusted_ed_visit_rate, fill = ces_category
  geom_boxplot() +
  labs(
    title = "Distribution of Asthma ED Rates by CES Category",
    x = "CES Category",
    y = "Age-adjusted ED Visit Rate (per 100,000)",
    caption = "Data: CalEnviroScreen 4.0 and HCAI Asthma ED Visits (2020)"
  ) +
  theme_minimal()
```




```r
ggplot(df_combined, aes(x = median_poverty, y = ed_visits_per_100k)) +
geom_point(color = "darkgreen", size = 3) +
geom_smooth(method = "lm", se = TRUE, color = "black") +
labs(
  title = "Asthma ED Rates vs. Median Poverty by County",
  x = "Median Poverty Rate (%)",
  y = "ED Visit Rate (per 100,000)",
  caption = "Data: CalEnviroScreen 4.0 and HCAI Asthma ED Visits (2020)"
) +
theme_minimal()
```

```
`geom_smooth()` using formula = 'y ~ x'
```



Interpretation: Counties with higher poverty rates often exhibit higher asthma ED visit rates, suggesting that socioeconomic factors may contribute independently to asthma burden.